Fiscal Year:	FY 2012	**Task Last Updated:**	FY 05/25/2012
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Project Title:	Rodent spine deconditioning after 30 days of microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology (2) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2007 Space Flight: Bion-M1 NNH07ZTT001N
Start Date:	08/01/2009	**End Date:**	07/31/2014
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
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Flight Program:	Bion/Shuttle		
Flight Assignment:	STS-131 ; Bion M1 NOTE: Extended to 7/31/2014 per A. Chu/ARC (Ed., 4/30/2014)		
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Grant/Contract No.:	NNX09AP11G		
Performance Goal No.:			

Performance Goal Text:

Task Description:

The vertebral bodies and flexible intervertebral discs of the spine are important, weight-bearing tissues that are adapted to gravitational stress. The absence of gravitational axial loads on the body during exposure to microgravity may disrupt normal spine physiology and function. During microgravity exposure, deconditioning of the intervertebral discs and spinal tissues poses a serious risk upon re-exposure to 1-G and exercise. Our recent data in identical twins exposed to 30-days bed rest indicate that compressibility increases in the lumbar spine and its intervertebral discs during 30-days simulated microgravity. Because intervertebral discs become more collapsible during simulated microgravity, the probability of disc prolapse or herniation upon re-exposure to gravity increases.
We hypothesize that nucleus pulposus swelling pressures in mouse spines will significantly decrease following 15- and 30-days of microgravity. In addition, we hypothesize that lack of gravitational loading on lumbar discs during microgravity will lead to extracellular matrix damage and significantly reduced disc compression performance following 15- and 30-days of microgravity. To test our hypotheses, we will harvest and test spines from ground-based control mice and flight mice exposed to 15 and 30-days of actual microgravity on STS-131 and Bion-M1. Spines will be frozen and transported to UCSD and UCSF for accepted methods of testing nucleus pulposus swelling pressure, lumbar disc dimensions, proteoglycan content, and spine biomechanics. We will compare the results of the present study to our peer-reviewed and published results for identical twins exposed to 30-days simulated microgravity and control mice exposed to 1-G. Our research will aid understanding of spinal deconditioning during 15- and 30-days microgravity and of the higher incidence of disc prolapse or herniation following re-exposure to 1-G with a long-term view to prevent such spinal deconditioning with exercise or other physiologic countermeasures.

Separately, evidence suggests headward fluid shifts alone influence bone alteration. However, most conclusions are drawn from human bed rest studies and hindlimb unloading of rats aimed to simulate actual weightlessness. As a member of NASA's Biospecimen Sharing Program, our lab acquired murine calvariae from a 15-day mission aboard Space Shuttle STS-131. In concordance with current theory, we hypothesized that compared to ground control mice, headward fluid shifts in mice exposed to microgravity will induce adaptive growth in bones that normally do not bear weight.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

To our knowledge, this study is the first to examine the effects of 15- and 30-days space flight on spinal deconditioning in mice. The vertebral bodies and flexible intervertebral discs are important, weight-bearing tissues that have adapted to gravitational stress. Our research will aid understanding of spinal deconditioning during 15- and 30-days microgravity and of the higher incidence of disc prolapse or herniation following re-exposure to 1-G with a long-term view to prevent such spinal deconditioning with exercise or other physiologic countermeasures. This research may aid understanding of spinal deconditioning during inactivity such as after spinal cord injury and bed rest in human patients on Earth. Separately, our studies of mouse skull calvaria demonstrate the influence of the fluid shift associated with microgravity on bone remodeling independent of skeletal loading. The findings may be considered in the development of strategies to counter bone loss in astronauts and patients confined to bed rest.

I. The Effects of Microgravity on Mice Lumbar Vertebral Bone Structure
INTRODUCTION: Extended exposure to microgravity during spaceflight has been associated with bone remodeling. Reduced mechanical loading, due to hypodynamia (decreased forces) and hypokinesia (fewer movements), was thought to lead to decrease in bone mineral density (BMD) during spaceflight. Previous studies have indeed reported decreased bone mass and vertebral BMD during long-term spaceflight. The use of micro-computed tomography (μCT) has recently been established to characterize deterioration of the micro-architecture of bone tissue, and to measure the BMD. With the support of NASA's mission STS-131 with the Space Shuttle Discovery, mice lumbar spines from microgravity conditions (15 days) were obtained. The purpose of this study was to assess the effect of microgravity on mice lumbar vertebral bone structure using μCT imaging.

METHODS:

Sample: Fifteen C57BL/6 mice were used, primarily for immunological research regarding the ability to activate protective immune responses during microgravity. The flight group (n=8, average weight 25 g) was sent to space on mission STS-131 with NASA's Animal Enclosure Module (AEM) and experienced microgravity for fifteen days. The AEM is a self-contained environment that holds up to eight adult mice and provides the mice with light, food, water, shelter, and ventilation. The control group (n=7) remained on Earth (Kennedy Space Center, FL) under the same conditions as the AEM (humidity, temperature, etc.). Upon arrival, the animals were euthanized, and the lumbar spines (L3 to L6) were harvested, frozen in liquid nitrogen, and transported frozen to the University of California, San Diego.
Micro-Computed Tomography: L3-L6 lumbar spines were imaged on a μCT scanner, Skyscan 1076.

RESULTS: Trabecular bone thickness in the flight group showed a trend lower values compared to the control group ($p=0.11$). Trabecular bone thickness at the caudal location in the flight group was significantly less than that of the control group ($p<0.05$), while trabecular bone thickness at the caudal location showed no difference. The flight group also showed a decrease of trabecular bone thickness at L6 compared to the control group ($p<0.05$), but not other levels.

DISCUSSION: In this study, the bone structure parameters in the flight group showed a trend of decrease compared to the control group. These decreases were more evident in caudal lumbar vertebra than in cranial vertebra. These results suggested the biomechanical condition of lumbar spine may have resulted in different responses to microgravity. The use of μCT imaging contributed to micro-architecture deterioration of trabecular bone in the lumbar vertebral body induced by a 15 day period of microgravity. This finding indicated that the loss of bone mass and mineralization may occur mainly in the center of the vertebral body under microgravity conditions.

II. The Effects of Microgravity on Mice Lumbar Disc MR Properties

INTRODUCTION: Extended exposure to microgravity during spaceflight has been associated with increased incidences of back pain in astronauts. In animals, signs of intervertebral disc (IVD) degeneration such as decreased proteoglycan content has been found. MRI is increasingly used for routine disc evaluation, and T2 quantification is sensitive to IVD hydration and proteoglycan content, as well as therapeutic repair of IVDs in animal models. The objective of this study

Task Progress:

was to determine the effects of 15-day spaceflight on MR properties of IVD in mice that experienced microgravity.

METHODS: Samples - Black congenic C57BL/6 mice (average weight of 25 g) were obtained from a non-treatment group at the NASA tissue share program. They were intended for immunological research regarding the ability to activate protective immune responses during microgravity. One group of mice was sent to space on mission STS-131 with NASA's Animal Enclosure Module (AEM) and experienced microgravity for 15 days (Flight; n=8). Another group (Control; n=8) stayed on Earth (Kennedy Space Center, FL) under the same conditions as the AEM (i.e., humidity and temperature). Upon arrival, the animals were euthanized, and the lumbar spines were harvested from each animal, frozen in liquid nitrogen, and transported frozen to the MRI facility. MRI Analysis – Spines were thawed, placed into plastic tubing with closed ends, and imaged using a 3T GE Signa HDx with a 1 cm diameter solenoid coil. RESULTS: In T2-weighted images, no discernible differences were found between the two groups. T2 values of the nucleus pulposus ranged from ~60 ms to 75 ms, and tended to be higher in the Control than the Flight animals. Statistically, the nucleus pulposus varied significantly with level ($p<0.001$) but not with microgravity and flight ($p=0.5$). T2 values of the nucleus pulposus were to be the lowest in L6/S1 and L5/6 compared to other levels (posthoc $p<0.05$). T2 values of the annulus fibrosis also tended to vary with level ($p=0.08$) but not with spaceflight ($p=0.3$), even though a trend of lower T2 values in the Flight group was apparent. In the annulus fibrosis, T2 values in the L1/2 was higher than those in the L6/S1 ($p<0.05$).

DISCUSSION: Although failing to reach statistical significance due to animal variability, a visible trend of decreasing T2 values in both annulus fibrosis and nucleus pulposus after spaceflight was observed. Additional samples may be needed to reduce animal variability, and longer duration flight may amplify the difference in the MR property. In bed rest studies, durations ~>10 weeks were needed to have a long-lasting effect on disc area.

SIGNIFICANCE: Quantitative MRI may be a sensitive tool for evaluation of disc degeneration associated with spaceflight.

III. Prolonged microgravity degenerates biomechanical properties of the murine caudal intervertebral discs

INTRODUCTION: Prolonged exposure to microgravity has adverse effects on the human intervertebral disc (IVD), causing low back pain during spaceflight and an increased incidence of post-spaceflight herniated nucleus pulposus (HNP). Cervical HNP incidence among an astronaut sample population was nearly 5-fold higher compared to a control population, while incidence of lumbar HNP was not significantly different. Theoretically, microgravity increases spinal injury risk by removing diurnal loading that expels IVD fluid that has been osmotically imbibed during bed rest. Discs freely swell throughout the entire flight, leading to increased disc height and spinal stiffness among astronauts. We propose that degenerative effects of microgravity will vary among spinal regions due to level-dependent differences in range of motion (ROM). Specifically, imposing larger ROM on microgravity stiffened spinal segments could cause accelerated degeneration and increased HNP risk immediately post-spaceflight. As part of the Biospecimen Sharing Program with NASA, we acquired murine lumbar spines and tails from STS-131, a 15-day shuttle mission. We hypothesize the caudal IVD will exhibit a greater difference in biomechanical properties between the spaceflight and ground control mice due to the greater ROM at caudal segments compared to lumbar segments.

METHODS:

Animals. Sixteen female, 16-week-old, wild-type C57BL/C mice were separated into two groups. The spaceflight group (n=8 mice) were exposed to microgravity for 15 days during NASA shuttle mission STS-131. The control group (n=8 mice) was maintained under normal gravitational loads on the ground. Following spaceflight, mice were sacrificed immediately post-landing. Eight caudal specimens (C9/C10) and sixteen lumbar specimens (L4/L5) were utilized for testing. Specimens were radiographed prior to dissection to attain a measure of physiologic disc height. Motion segments were harvested and then dissected by removing extraneous soft-tissue, as well as, posterior elements from the lumbar specimens.

Mechanical Testing: The motion segments were fixed to the testing apparatus (Bose Electroforce 3200) and radiographed again to measure the disc cross-sectional area and initial disc height at the start of the test. Testing included five cycles of compressive creep loading at 0.5 MPa for 20 minutes with 40 minutes recovery at 0.1 MPa. The first four cycles were used for preconditioning and the last cycle was used for data collection and analysis.

RESULTS: Seven of the eight caudal IVDs contributed usable data. Within that set, the spaceflight specimens (n=4) demonstrated 32% ($p=0.012$) decrease in physiologic disc height and 70% decrease in strain-dependence of swelling pressure, D ($p=0.02$) compared to the ground control mice (n=3). D was moderately correlated with disc height ($R2=0.56$, $p=0.0013$).

DISCUSSION: The spaceflight caudal IVDs showed significant decreases in physiological disc height and strain dependence (D) compared to the corresponding ground controls. Decreases in either variable are indicative of disc degeneration, indicating that microgravity has an injurious effect on the caudal IVD. By contrast, differences were not noted for lumbar IVDs. Lumbar spine movement is constrained by the disc, the ligaments, and the facet joints. In the tail however, there are no facets, and as a result, the range of motion is larger and constrained principally by the disc. Consequently, when supra-physiologic nuclear swelling from prolonged weightlessness increases disc bending stiffness, this can be more problematical in the tail. This potentially deleterious interaction between prolonged weightlessness and spine range of motion may underlie the increased cervical versus lumbar disc herniation rates observed in crewmembers. In humans, the cervical range of motion is significantly greater than the in the lumbar spine for both flexion/extension and axial rotation. The incidence of cervical HNP is 21.4-times higher after spaceflight, while the lumbar HNP was not significantly higher. If these results are confirmed, they suggest countermeasure efforts should focus on protecting spinal movement upon return to Earth's gravity while nucleus swelling can acclimate to terresterial conditions.

SIGNIFICANCE: This study shows that prolonged microgravity could have degenerative effects on IVDs at levels with a greater range of motion. Results could influence potential countermeasures to be used during spaceflight to reduce the occurrence of HNP.

IV. Post-spaceflight recovery of biomechanical properties in murine caudal intervertebral discs

INTRODUCTION: Prolonged exposure to microgravity during spaceflight adversely affects human intervertebral disc (IVD) increasing the incidence of disc herniation post-spaceflight. The risk of disc herniation is highest during the first year and subsides over time, but persisting at high risk for up to 3 years post-spaceflight. We hypothesize that the biomechanical properties of the IVD undergo degeneration during prolonged exposure to microgravity and that

biomechanical properties gradually recover during acclimation to gravitational loads post-flight. We will test this hypothesis by comparing the creep compressive properties of caudal IVDs belonging to murine subjects from STS-133 to control IVDs.

METHODS: Sixteen female, 13-week-old, wild-type BALB/c mice were divided into the spaceflight group (n = 6 mice) and the control group (n =10 mice). Mice were sacrificed at three different post-landing time points: +1 day of recovery (spaceflight, n=2; control, n=3), +5 days of recovery (spaceflight, n=2; control, n=4), and +7 days of recovery (spaceflight, n=2; control, n=3). Discs were subjected to five cycles of compressive creep loading at 0.5 MPa for 20 min with 40 min recovery. The displacement vs. time data were fit to a fluid transport model to calculate tissue permeability (K), time-dependence of annular strain (G), and strain-dependence of nucleus swelling pressure (D).

RESULTS: The spaceflight group exhibited a 12% decrease in physiological disc height (p=0.0987) and a 21 % decrease in D (p = 0.1346) compared to controls. We found no significant effect of post-flight recovery time on the biomechanical properties. We observed no significant differences for the creep parameters K and G.

DISCUSSION: We found that prolonged exposure to microgravity tended to diminish the biomechanical properties of the murine caudal IVD. This supports the claim that microgravity induces degenerative changes to the IVD; however it remains unclear whether these changes are reversible upon re-acclimation to gravitational loads. The lack of significant differences in biomechanical properties between the three recovery periods could be a result of the small sample size (n=2 for each spaceflight mice time point). It may also reflect the short duration of recovery time.

V. Fifteen Days Microgravity Causes Growth in Murine Calvaria on STS-131

INTRODUCTION: Bone remodeling occurs in spaceflight as a response to skeletal unloading and headward fluid shifts. While unloading causes significant loss of bone mass and density, increased blood and interstitial fluid flows may elicit an opposite effect. In bones that normally do not bear weight, we hypothesized that the headward fluid shift accompanying microgravity can induce adaptive growth in skull-bone structure. We investigate this relationship by comparing the calvariae of mice exposed to 15-days microgravity compared to ground-control mice.

METHODS: Seven of 15 wild-type C57BL/6 mice littermates were randomly chosen to 15 days of microgravity on the STS-131 mission, and the remaining eight mice served as ground controls. Upon mission completion, all calvariae were imaged on a micro-Computed Tomography scanner (Skyscan 1076). A standardized rectangular volume (18mm3) was placed on the parietal bones of each calvaria for analysis. The parietal bones were chosen to represent the calvaria due to their central locations on the skullcap. Three parameters that characterized bone growth were determined: bone volume (BV; expressed as % of the standardized rectangular volume), cross-sectional thickness (CsTh; expressed in mm), and tissue mineral density (TMD; expressed in g/cc).

RESULTS: Increases of spaceflight group (SF) over ground control group (GC) were seen in all three parameters. BV was increased ($p<0.05$) from a mean of 9.64% ±0.79 in the GC group to 10.48% ±0.47 in the SF group, with the relative difference being +8.7%. CsTh showed a trend of increase (p=0.12) from a mean of 0.099mm ±0.006 in the GC group to 0.104mm ±0.005 in the SF group, with the relative difference being +5.1%. TMD was similar between the two groups (p=0.31) with a mean of 0.878g/cc ±0.029 in the GC group, 0.893g/cc ± 0.028 in the SF group, and a relative difference of 1.8%.

DISCUSSION: Our results suggest that head-ward fluid shifts associated with microgravity initiate adaptive growth in bones that don’t normally bear weight. The findings also indicate the capability of a fluid shift to affect bone remodeling independent of skeletal loading.

Bibliography Type:	Description: (Last Updated: 08/25/2025)
Abstracts for Journals and Proceedings	Bailey JF, Cheng KK, Hargens AR, Masuda K, Lotz JC. "Prolonged microgravity degenerates biomechanical properties of the murine caudal intervertebral discs." Presented at 2012 Annual Meeting of the Orthopaedic Research Society, San Francisco, CA , February 4-7, 2012. Transactions of the 58th Annual Meeting of the Orthopaedic Research Society, 2012. , Feb-2012
Abstracts for Journals and Proceedings	Lotz J, Hargens AR. "Spines in Space: Microgravity Effects on Spinal Discs." Presented at 2012 Annual Meeting of the Orthopaedic Research Society, San Francisco, CA , February 4-7, 2012. Transactions of the 58th Annual Meeting of the Orthopaedic Research Society, 2012. , Feb-2012
Abstracts for Journals and Proceedings	Bae WC, Chen N, Yamaguchi T, Bydder GM, Hargens AR, Masuda K. "Effects of Space Flight on T2 MR Properties of Mouse Intervertebral Discs." Presented at 2012 Annual Meeting of the Orthopaedic Research Society, San Francisco, CA , February 4-7, 2012. Transactions of the 58th Annual Meeting of the Orthopaedic Research Society, 2012. , Feb-2012
Abstracts for Journals and Proceedings	Bailey JF, Cheng KK, Hargens AR, Masuda K, Lotz JC. "Post-spaceflight recovery of biomechanical properties in murine caudal intervertebral discs." Presented at Orthopaedic Research Society, San Francisco, CA Feb 5, 2012., Transactions of the 58th Annual Meeting of the Orthopaedic Research Society, 2012. , Feb-2012
Abstracts for Journals and Proceedings	Bailey JF, Hargens AR, Cheng KK, Masuda K, Lotz JC. "Prolonged microgravity degenerates biomechanical properties of the murine caudal intervertebral discs." Presented at 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. Program and abstracts. 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. p. 79. , Nov-2011
Abstracts for Journals and Proceedings	Bailey JF, Hargens AR, Cheng KK, Masuda K, Lotz JC. "Post-spaceflight recovery of biomechanical properties in murine caudal intervertebral discs." Presented at 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. Program and abstracts. 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. p. 79-80. , Nov-2011